



— DEFENSE & NATIONAL SECURITY

IGNITING DEFENSE INNOVATION THROUGH SCIENCE-POWERED SOLUTIONS.

SCIENCE AND TECHNOLOGY WITH PASSION

Engineering excellence trusted on NASA's flagship missions, now enabling defense and national security. SSAI is a science-powered science, technology, and engineering company. We bring zero-trust cybersecurity, AI/ML, a cleared Mil-spec manufacturing base, deep-space operations, and instrument engineering to the defense enterprise. Built for national security, focused on operational dominance.

1977

FOUNDED · 45+ YEARS

575+

SCIENTISTS,
ENGINEERS & IT
PROFESSIONALS

TS FCL

TOP SECRET FACILITY
CLEARANCE

42K

SQ FT CLEARED
MANUFACTURING &
MACHINE SHOP

100+

MAJOR INSTRUMENTS,
MODELS & MISSIONS

3

WHOLLY-OWNED
SUBSIDIARIES

DEFENSE CAPABILITIES

BUILT FOR NATIONAL SECURITY, FOCUSED ON OPERATIONAL DOMINANCE.

A cleared, vertically-integrated provider spanning digital defense, cleared manufacturing, space operations, and mission-grade engineering.

- 01**
CYBERSECURITY & DIGITAL DEFENSE
Advanced IT and cybersecurity built on zero-trust architectures, with AI/ML for anomaly detection, autonomy, and cross-agency support. Our CREST framework (Coupled Reusable Earth System Tensor) is an AI-enabled Python framework for operational digital twins. Custom AI engineered for the mission, from sensor to decision.
- 02**
SATELLITE & SPACE SYSTEMS MANUFACTURING
A 32,000 sq ft cleared electronics manufacturing facility (PCB design, fab, and test; FPGA board repair; ESD-protected floor; three clean rooms; cleared personnel) and a 10,000 sq ft Mil-spec machine shop (Mil-I-45208A, Mil-std-45662; CNC, lathes, presses, mills; certified aerospace machinists) under two miles from NASA Goddard.
- 03**
SPACE OPERATIONS
Deep-space and space-weather operations: 24/7 operational modeling, deep-space instrument development and calibration, trajectory design and atmospheric-entry modeling, and mission operations from pre-launch through on-orbit. Space domain expertise that transfers directly to defense and resilient space.
- 04**
MISSION-GRADE ENGINEERING
Systems engineering, requirements and V&V, avionics and FPGA design, software engineering (CMMI L3), EEE parts processing, and system integration and testing. AS9100, ITAR registered, J-STD-001F workmanship. The engineering rigor that flies NASA flagships, applied to defense hardware.
- 05**
ENVIRONMENTAL INTELLIGENCE
Geospatial and remote-sensing analytics that turn raw sensor data into actionable intelligence: data assimilation, remote sensing, weather and space-weather forecasting, and multi-source fusion for ISR and operational decision advantage.

MISSION ALIGNMENT

MAPPED TO THE DEFENSE PORTFOLIOS THAT MATTER.

AI / ML

CREST digital twins, anomaly detection, custom AI.

AUTONOMY

Autonomous sensing, modeling, and decision support.

CYBER & TELECOM

Zero-trust architectures, secure IT and comms.

SPACE

Space operations, instruments, resilient space.

EMERGING TECHNOLOGY

FPGA, digital twin, advanced analytics.

ENERGY

Environmental modeling and resource intelligence.

HUMAN SYSTEMS

Decision-grade data products and visualization.

Top Secret Facility Clearance

UEI PJ2SP76D3HJ3

CAGE 5S009

NAICS 541715 (small) / 541330

ITAR Registered · WOSB

AS9100 · CMMI L3 · ISO 9001

DIFFERENTIATORS

CLEARED, CERTIFIED, CORRIDOR-BASED.

- **Top Secret Facility Clearance.** Cleared facility and cleared personnel, ITAR registered, ready for classified work.
- **Cleared, Mil-spec manufacturing base.** 42,000 sq ft of in-house electronics manufacturing and precision machining under two miles from NASA Goddard.
- **In the D.C. defense corridor.** Minutes from NASA Goddard and the Washington defense community, where requirements are written and missions are run.
- **Aerospace-grade engineering.** AS9100, CMMI L3 for Development, ISO 9001:2015, EEE parts processing, and J-STD-001F workmanship.

Top Secret FCL

ITAR Registered

AS9100 Aerospace Quality

CMMI L3 Development

ISO 9001:2015

SUBSIDIARIES

A SINGLE, TRUSTED SOURCE.

Three wholly-owned subsidiaries deliver mission-ready capability in house.

ADVANCED MISSION PARTNERS

Electronics manufacturing, engineering services, EEE expertise, and precision manufacturing.

ELUCIDATION CONCEPTS, LLC

Cybersecurity solutions, systems engineering, system integration testing, and test & evaluation.

SIGMA ENGINEERING & FABRICATION, INC.

Precision machining, manufacturing, material expertise, and quality at speed.

PAST PERFORMANCE

DELIVERING MISSION-CRITICAL SCIENCE AND ENGINEERING.

ESSES III	NASA Goddard, Engineering Technology Directorate (CPAF). Study, design, development, fabrication, integration, testing, verification, and operations of spaceflight, airborne, and ground hardware and software. 11/2018 to 04/2025.	\$685M SSAI TO CONFIRM
HBG	Hydrospheres, Biospheres and Geophysics. NASA Goddard, Earth Sciences Division (CPFF). Scientific research across space, airborne, in-situ, and laboratory observations. 04/2020 to 11/2025.	\$425M SSAI TO CONFIRM
STARSS III	Science, Technology and Research Support Services. NASA Langley, Science Directorate (CPFF). Earth atmosphere and climate, extending to other planetary atmospheres. 12/2016 to 05/2023.	\$307M SSAI TO CONFIRM
SAMDA	Support for Atmospheric Modeling and Data Assimilation. NASA Goddard, Global Modeling and Assimilation Office (CPFF). 03/2017 to 11/2025.	\$298M SSAI TO CONFIRM
HBS	Hydrospheric and Biospheric Sciences. NASA Goddard, Earth Sciences Division (CPFF). 12/2014 to 05/2020.	\$249M SSAI TO CONFIRM

Flagship missions: OCI / PACE · Roman Space Telescope · Lucy · Mars Sample Return / CCRS · Dragonfly · GeoXO · DAVINCI+ · TSIS-2 · JWST.

CORPORATE DATA

THE ESSENTIALS.

UEI	PJ2SP76D3HJ3 CONFIRM	CAGE	5S009 CONFIRM
NAICS	541715 (small) / 541330	BUSINESS STATUS	WOSB, SBA 2024 CONFIRM
GSA MAS	47QRAA23D0014 CONFIRM	IDIQ VEHICLE	OASIS Small Business CONFIRM
SECURITY	Top Secret FCL, ITAR Registered	QUALITY & MATURITY	AS9100, CMMI L3, ISO 9001:2015
FOUNDED	1977	HEADQUARTERS	Lanham, MD

BUSINESS DEVELOPMENT

BD@ssaihq.com
(301) 867-2000

HEADQUARTERS

The Aerospace Building
10210 Greenbelt Rd, Suite 600
Lanham, MD 20706

LOCATION ADVANTAGE

Minutes from NASA Goddard, in the Washington D.C. defense corridor.

Procurement identifiers and contract values marked "confirm" / "SSAI to confirm" are pending SSAI sign-off before external release.